■ Magic & Shell Commands in Jupyter Notebooks

Quick Reference Summary for Data Science & Analysis

Overview

Magic Commands are special IPython features designed to simplify data analysis and system tasks. They come in two types: **line magics** (single %) for one line, and **cell magics** (double %%) for the whole cell.

Performance Measurement Magics

Command	Type	Description
%time	Line	Measures execution time of a single-line statement (single run).
%%time	Cell	Measures execution time for all code in a cell (single run).
%timeit	Line	Average execution time over multiple runs (single line).
%%timeit	Cell	Average execution time over multiple runs (entire cell).

Script Execution & History Magics

%run filename.py	Runs a Python script inside IPython.
%history	Displays command history. Use -n to show last n commands.
%recall <line_no>	Re-executes a command by line number or range.

Variable & Help Magics

%who	Lists all variables defined in the current session.
%lsmagic	Lists all available magics.
%magic	Shows detailed help for all magic commands.
%quickref	Displays a quick reference guide of common magics.

Shell Commands in Jupyter

Shell commands are standard terminal commands that can be used in Jupyter by prefixing them with **!**. They can also have magic equivalents (e.g., %ls, %pwd).

Command	Example	Description
!ls	!ls	Lists files and directories.
!pwd	!pwd	Prints current working directory.
!cat file.txt	!cat data.csv	Prints file contents.
!head file.csv	!head data.csv	Shows first 10 lines of a file.

Quick Tips

• Use %lsmagic to explore all magics.
• Combine Python and shell commands for flexible workflows.
• Magics are ideal for performance checks, debugging, and exploration.
• Use list comprehensions with %timeit for efficiency tests.

Example Workflow

```
%timeit [x**2 for x in range(1000)]
%who
!ls
%run my_script.py
```